AuraSense: An AI-Driven Sensor Fusion Retrofit Kit for Smart Tea Drying and Quality Optimization in Legacy Factories

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Abstract

The tea industry in Assam is currently facing severe economic pressures due to rising thermal fuel costs and unpredictable climate variations affecting crop moisture. Drying, the most energy-intensive process in tea manufacturing, consumes up to 80% of a factory's thermal energy. Currently, operations rely on reactive Proportional-Integral-Derivative (PID) controllers and subjective human sensory evaluation, leading to significant energy waste and inconsistent product quality.

This project proposes **AuraSense**, a low-CAPEX, AI-driven retrofit kit designed to upgrade legacy tea dryers (ECP and FBD). By integrating a multi-modal sensor fusion architecture—comprising an 8-channel Metal Oxide Semiconductor (MOS) Electronic Nose, a 3-zone thermocouple array, an exhaust humidity sensor, and a Computer Vision colorimeter—with a mathematically rigorous Digital Twin, the system captures objective, real-time quality metrics. A Reinforcement Learning (RL) agent trained with Proximal Policy Optimization (PPO) acts as an operator Co-Pilot, predicting the optimal thermodynamic trajectory to minimize Specific Energy Consumption (SEC) while maximizing the biochemical flavor profile (Theaflavins and Pyrazines).

The complete system has been implemented as a full-stack web application: a Python/FastAPI backend hosting the physics engine and RL agent, communicating via WebSocket telemetry to a React Three Fiber frontend rendering a live 3D Digital Twin with real-time analytics. This report presents the theoretical framework, system architecture, implementation methodology, simulation results comparing RL vs. PID control, and the techno-economic feasibility of deploying AuraSense via a Hardware-as-a-Service (HaaS) business model.

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Chapter 1

Introduction

1.1 Motivation

The production of black tea is a highly sensitive biochemical process where the final organoleptic profile—and consequently, the auction price—is dictated by the precision of the drying phase. In Assam, the industry relies heavily on Bought Leaf Factories (BLFs) operating on razor-thin margins. The motivation for this project is to democratize Industry 4.0 technologies. BLFs lack the capital to invest in modern, intelligent dryers; therefore, an affordable, non-disruptive retrofit solution is urgently required to improve profit margins and sustainability.

The specific challenges motivating AuraSense include:

1. **Transport Delay:** The thermal lag between burner adjustment and the tea bed actually reaching the new temperature (measured as $\tau \approx 2$ minutes in FBD systems) means reactive controllers always act too late.
2. **Stewing Risk:** When bed temperature remains below 60°C while moisture exceeds 40%, Polyphenol Oxidase (PPO) continues unchecked oxidation, producing a flat, dull liquor—the dreaded "stewing" defect.
3. **Subjective Quality Assessment:** Factory tasters are retiring without successors, and human sensory evaluation is inherently inconsistent across shifts.
4. **Energy Waste:** Temperature overshoot and under-drying followed by re-drying cycles waste 15–20% of thermal fuel.

1.2 Problem Statement

Current tea drying operations suffer from a "black box" approach. Legacy dryers rely on reactive controllers that exhibit significant transport delay (dead time). By the time an error is detected at the exhaust, the batch is already compromised—either case-hardened due to excessive heat or stewed due to insufficient heat. This inefficiency wastes 15–20% of thermal fuel (coal/gas) and relies entirely on a shrinking workforce of aging human experts for quality assessment.

1.3 Project Objectives

- **Hardware Integration:** To design a low-cost, multi-modal retrofit sensor node comprising an industrial RGB camera, an MQ-series gas sensor array (8 channels), a 3-zone thermocouple array, and an exhaust humidity sensor.
- **Digital Twin Simulation:** To develop a thermodynamic physics engine in Python that accurately models heat/mass transfer (Page Model), enzyme kinetics (Arrhenius), Maillard flavor genesis, and CIELAB colorimetric evolution.

- **AI/ML Control:** To train a Reinforcement Learning (RL) agent using Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) algorithms, utilizing sensor fusion data to optimize the drying trajectory.
- **Real-Time Visualization:** To build a 3D Digital Twin frontend using React Three Fiber with live WebSocket telemetry, enabling factory operators to monitor drying in real time.
- **Entrepreneurial Strategy:** To formulate a viable Hardware-as-a-Service (HaaS) business model and calculate unit economics for scaling the technology across the Assam tea industry.

1.4 Scope and Deliverables

The project delivers a complete, deployable software system consisting of:

- A Python physics engine with 4 sub-models (moisture, enzyme, flavor, color) validated by 37 automated unit tests.
- A synthetic 4-sensor simulation layer (e-nose, thermocouple, humidity, colorimeter) producing realistic noisy signals.
- A FastAPI backend with REST API and WebSocket streaming at 10 Hz.
- A Gymnasium-compatible RL environment with a multi-objective reward function (7 components).
- A React/Three.js frontend with 3D visualization, live analytics charts, and simulation controls.
- A Dockerfile for one-command deployment.

Chapter 2

Literature Review

The integration of Machine Learning and sensor technologies in agricultural processing is a rapidly expanding field.

2.1 Electronic Noses for Aroma Profiling

Portable electronic noses (E-Noses) equipped with MOS sensor arrays have been demonstrated to objectively assess tea aroma, bypassing subjective human testing. Bhattacharyya et al. [1] showed that e-noses coupled with Random Forest classifiers can accurately classify tea grades by monitoring volatile organic compounds (VOCs) emitted during processing, achieving classification accuracies exceeding 95%. The TGS-series MOS sensors (TGS 2600, TGS 2602, TGS 2610) are particularly suitable due to their sensitivity to alcohols, aldehydes, and heterocyclic compounds at the ppm level.

2.2 Computer Vision in Quality Assessment

Machine vision systems have been successfully developed to evaluate the fermentation and drying quality of black tea. Gill et al. [2] demonstrated that by extracting RGB, HSV, and CIELAB features from leaf-bed images, chemometric models can correlate color characteristics with chemical compositions (Theaflavins/Thearubigins ratio), achieving near-perfect discrimination accuracy ($R^2 > 0.96$). The CIELAB color space is preferred because it is perceptually uniform—equal numerical differences correspond to equal perceived color differences.

2.3 Model Predictive Control and AI in Drying

Deep Learning Model Predictive Control (DL-MPC) has been explored by Sturm et al. [3] to address the non-linear dynamics of convective drying. Their work demonstrates that shifting from traditional PID to predictive control reduces energy consumption by 12–18% while preserving heat-sensitive antioxidants. More recently, Reinforcement Learning has emerged as a model-free alternative that can learn optimal policies directly from interaction, without requiring an explicit plant model [4].

2.4 Digital Twins in Manufacturing

The Digital Twin paradigm—a virtual replica of a physical asset updated in real time with sensor data—has gained significant traction in Industry 4.0. Tao et al. [5] formalized the five-dimensional Digital Twin model comprising Physical Entity, Virtual Entity, Connection, Twin Data, and Services. In food processing, Digital Twins enable non-destructive quality prediction and predictive maintenance, reducing downtime by up to 30%.

2.5 Thin-Layer Drying Models

The Page Model, an empirical modification of Newton's law of cooling applied to mass transfer, remains the most widely used thin-layer drying equation for agricultural products. Panchariya et al. [6] fitted the Page constants for CTC black tea ($k = 0.12$ min^{-1}, $n = 1.04$) at an air velocity of 1.0 m/s and inlet temperature of 100°C, reporting an $R^2 > 0.99$ fit to experimental data.

Chapter 3

Theoretical Framework & Process Kinetics

To accurately represent the thermochemical dynamics of tea drying without physical destructive testing, the AuraSense Digital Twin employs a mathematical physics engine based on first-principles thermodynamics and Arrhenius reaction kinetics. All sub-models are implemented in Python and coupled through a unified time-stepping driver (`dryer.py`).

3.1 Mass Transfer (Drying Kinetics)

The core moisture removal process is simulated using the empirical Page Model for thin-layer drying. The Moisture Ratio (MR) at any given time t is calculated as:

$$MR(t) = \exp(-k_{\text{eff}} \cdot t^n) \tag{3.1}$$

where the absolute moisture content is recovered via:

$$M(t) = M_e + (M_0 - M_e) \cdot MR(t) \tag{3.2}$$

The effective drying constant k_{eff} is modulated by inlet temperature and airflow to reflect real FBD behavior:

$$k_{\text{eff}} = k_{\text{base}} \cdot \left(\frac{T_{\text{inlet}}}{100}\right)^{1.5} \cdot v_{\text{air}}^{0.6} \tag{3.3}$$

This Arrhenius-like scaling ensures that hotter air and higher airflow accelerate drying. The instantaneous drying rate is obtained analytically:

$$\frac{dM}{dt} = -(M_0 - M_e) \cdot k_{\text{eff}} \cdot n \cdot t^{(n-1)} \cdot \exp(-k_{\text{eff}} \cdot t^n) \tag{3.4}$$

The fitted parameters used in the simulation are listed in Table 3.1.

Table 3.1: Page Model Parameters for CTC Black Tea

Parameter	**Symbol**	**Value**	**Unit**
Drying constant	k	0.12	min^{-1}
Page exponent	n	1.04	—
Initial moisture	M_0	0.70	wet-basis fraction
Equilibrium moisture	M_e	0.03	wet-basis fraction
Ambient temperature	T_{amb}	28	°C
Default inlet temperature	T_{inlet}	100	°C
Default airflow	v_{air}	1.0	normalised (0–1)

3.2 Thermal Dynamics (Bed Temperature Lag)

The tea bed does not instantaneously reach the inlet air temperature. This transport delay is modeled as a first-order thermal lag:

$$\frac{dT_{\text{bed}}}{dt} = \frac{T_{\text{inlet}} - T_{\text{bed}}}{\tau} \tag{3.5}$$

where $\tau = 2.0$ minutes is the bed temperature lag time constant. This models the real-world delay between burner adjustment and the tea bed responding, and is the primary reason PID controllers perform poorly—they react *after* the delay, causing oscillation and overshoot.

3.3 Biochemical Degradation (Enzyme Inactivation)

To prevent the "stewing" defect, Polyphenol Oxidase (PPO) must be thermally inactivated rapidly. This is modeled as a temperature-dependent first-order kinetic degradation:

$$\frac{d[\text{PPO}]}{dt} = -k_{\text{denat}} \cdot [\text{PPO}] \tag{3.6}$$

The rate constant k_{denat} is governed by the Arrhenius Equation:

$$k_{\text{denat}} = A \cdot \exp\left(-\frac{E_a}{R \cdot T_{\text{bed}}}\right) \tag{3.7}$$

where $A = 1.0 \times 10^7$ s^{-1} is the pre-exponential factor, $E_a = 45{,}000$ J/mol is the activation energy for PPO denaturation, $R = 8.314$ J/(mol·K) is the universal gas constant, and T_{bed} is in Kelvin. Integration is performed using explicit Euler with clamping to $[0, 1]$.

3.3.1 Stewing Detection

A `StewingTracker` module monitors the dangerous zone where $T_{\text{bed}} < 60°\text{C}$ and $M > 0.40$. When cumulative time in this zone exceeds 5 minutes, a penalty flag is triggered. This flag feeds directly into the RL reward function and the UI warning system.

3.4 Flavor Genesis (The Maillard Reaction)

The formation of Pyrazines (roasty/toasty flavor compounds) via the Maillard reaction between amino acids and reducing sugars is simulated using zero-order kinetics:

$$\frac{d[\text{MRP}]}{dt} = \begin{cases} k_{\text{maillard}} & \text{if } T_{\text{bed}} \geq 110°\text{C and } M \leq 0.10 \\ 0 & \text{otherwise} \end{cases} \tag{3.8}$$

where $k_{\text{maillard}} = 0.08$ μg/(g·min). The zero-order assumption is valid because both substrates (amino acids, reducing sugars) are in large excess relative to the trace quantities of pyrazines produced.

3.5 Colorimetric Kinetics (CIELAB Model)

The visual transformation from greenish-copper (fresh rolled CTC leaf) to dark brown/black (dried tea) is modeled across all three CIELAB channels:

$$L^*(t) = L^*_{\min} + (L^*_0 - L^*_{\min}) \cdot \exp(-\lambda_L \cdot f_T \cdot t) \tag{3.9}$$

$$a^*(t) = a^*_f - (a^*_f - a^*_0) \cdot \exp(-\lambda_a \cdot f_T \cdot t) \tag{3.10}$$

$$b^*(t) = b^*_f + (b^*_0 - b^*_f) \cdot \exp(-\lambda_b \cdot f_T \cdot t) \tag{3.11}$$

where the temperature-dependent acceleration factor is:

$$f_T = \text{clamp}\left[\left(\frac{T_{\text{bed}}}{100}\right)^{1.2},\ 0.5,\ 2.0\right] \tag{3.12}$$

Table 3.2: CIELAB Color Model Parameters

Channel	**Initial**	**Final**	**λ (min^{-1})**	**Interpretation**
L^* (Lightness)	52	22	0.08	Darkening
a^* (Green→Red)	-8	12	0.07	Browning
b^* (Yellow→Brown)	25	10	0.06	Yellow loss

A full CIELAB → sRGB conversion pipeline (via XYZ tristimulus, D65 illuminant) is implemented so the frontend can directly color 3D meshes from the physics state.

Chapter 4

System Architecture

The AuraSense system is organized as a three-layer architecture: Perception (sensors), Application (Digital Twin + UI), and Control (RL agent). Figure 4.1 illustrates the data flow across all layers.

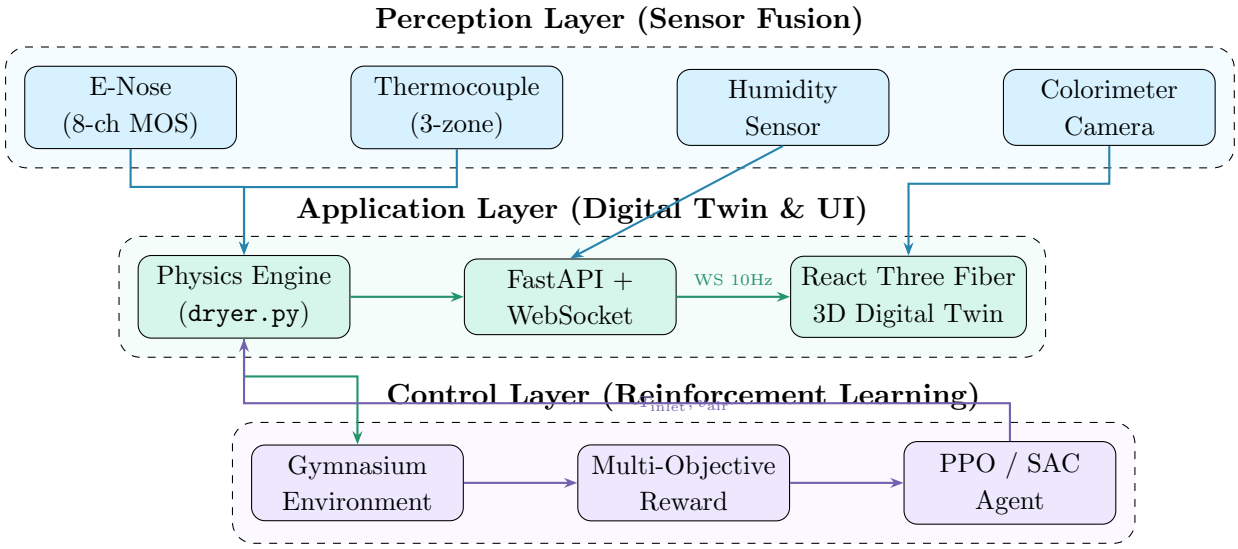

Figure 4.1: AuraSense three-layer system architecture showing data flow from sensors through the Digital Twin to the RL controller and back.

4.1 The Perception Layer (Hardware Retrofit)

The physical retrofit kit consists of four sensor subsystems, all simulated in `sensors.py` with realistic noise, drift, lag, and spatial variation:

1. **Electronic Nose (8-channel MOS array):** Eight virtual sensors modeled after the TGS 2600/2602/2610 family, each with a unique sensitivity profile:
   - Channels weighted toward moisture, enzyme-activity, pyrazine, and temperature in varying proportions.
   - Baseline voltage: 0.4 V; saturation: 4.5 V; Gaussian noise $\sigma = 0.02$ V; baseline drift: 0.001 V/min.
   - A composite VOC index (0–100 scale) is computed from the weighted mean.
2. **Thermocouple Array (3-zone):** Inlet, bed, and exhaust temperature measurements with $\sigma = 0.5$°C noise. The exhaust reading is derived from the bed temperature via $T_{\text{exhaust}} = T_{\text{bed}} - 15$°C with an additional first-order lag ($\tau = 1.5$ min).
3. **Humidity Sensor:** Exhaust-side relative humidity derived from the moisture mass balance. Ambient RH is set to 75% (Assam monsoon conditions), with $\sigma = 1.5\%$ RH noise. Dewpoint is computed via the Magnus approximation.
4. **Colorimeter Camera:** A top-down CIELAB measurement with $\sigma_L = 0.8$, $\sigma_{a,b} = 0.5$. Five spatial sample points across the bed produce a uniformity metric (0–1) detecting

localized case hardening.

All sensor readings are aggregated into a flat 17-dimensional observation vector for the RL agent: $[8 \times V_{\text{e-nose}},\ \text{VOC}_{\text{idx}},\ T_{\text{inlet}},\ T_{\text{bed}},\ T_{\text{exhaust}},\ \text{RH},\ L^*,\ a^*,\ b^*,\ \text{uniformity}]$.

4.2 The Application Layer (Digital Twin and UI)

4.2.1 Backend: FastAPI + WebSocket Streaming

The backend is built on **FastAPI** (Python 3.11) with an asynchronous lifecycle manager. Key components include:

- `SimManager` — Singleton simulation controller managing start/pause/stop/reset lifecycle, speed multipliers (1×–50×), and control input forwarding.
- `REST API` (`/api/`) — Endpoints for `GET /status`, `POST /start`, `POST /pause`, `POST /stop`, `POST /reset`, `POST /controls`, and `POST /speed`.
- `WebSocket` (`/ws/telemetry`) — Bidirectional channel streaming the full dryer state (including all sensor readings) at up to 10 Hz. Clients send JSON commands; the server pushes state snapshots.
- `CORS Middleware` — Configured to accept all origins during development, with header forwarding for credentials.

4.2.2 Frontend: React Three Fiber + Recharts

The frontend is a single-page application built with **Vite**, **React**, **Tailwind CSS v4**, and **Framer Motion**. It is organized into three pages accessible via an animated sidebar:

1. **Home Page** — Hero section with floating particle animation, feature cards (Digital Twin, RL Agent, Live Analytics, Sensor Fusion), live stat counters with spring-physics animations, and an architecture flow diagram.
2. **Digital Twin Page** — Split-screen layout:
   - *Left panel (55%):* A full 3D WebGL scene rendered via `@react-three/fiber` containing a detailed Fluidized Bed Dryer model with:
     - 500 tea particles (icosahedron geometry) that change color in real time based on the $L^*a^*b^*$ state via `lerp()`.
     - 120 heat-haze particles, 60 steam exhaust particles.
     - Structural details: flanges with 12 bolts each, support legs, perforated distributor plate, transparent glass cylinder.
     - 3D floating text labels, contact shadows, infinite grid floor.
   - *Right panel (45%):* Scrollable live analytics showing 6 stat cards (moisture, temperatures, airflow, enzyme, pyrazine), three Recharts time-series (moisture area chart, temperature line chart, quality line chart), sensor readouts (8-channel e-nose grid, thermocouple/humidity values), quality score card with color swatch, and 6 SVG arc gauges.

   A collapsible simulation control panel floats over the 3D scene with transport buttons, 6 speed presets, and inlet temperature / airflow sliders.

3. **Controls Page** — Full-page simulation control center with large sliders for inlet temperature (80–130°C) and airflow (20–100%), a mini 3D preview, and a live readout panel.

The sidebar uses Framer Motion `layoutId` for smooth animated tab-indicator transitions. All pages use staggered `fadeUp` entrance animations with spring physics. Page transitions use `AnimatePresence` with opacity/scale/translate animations (350 ms cubic-bezier ease).

4.3 The Control Layer (Reinforcement Learning Agent)

4.3.1 Gymnasium Environment

The FBD physics engine is wrapped in a standard **OpenAI Gymnasium** environment (`TeaDryerEnv`):

- **Observation Space:** $\mathbb{R}^{19}$, all normalized to $[0, 1]$:

$$\mathbf{o} = \left[M,\ \frac{T_{\text{bed}}}{150},\ \frac{T_{\text{in}}}{150},\ C_{\text{enz}},\ \frac{P}{5},\ \frac{L^*}{60},\ \frac{a^*+10}{25},\ \frac{b^*}{30},\ \dot{M},\ \frac{t}{t_{\text{dur}}},\ s_{\text{stew}},\ \frac{\mathbf{v}_{\text{e-nose}}}{4.5} \right] \tag{4.1}$$

- **Action Space:** $\mathbb{R}^2 \in [-1, 1]$, mapped to physical controls:

$$T_{\text{inlet}} = 25 \cdot a_0 + 105 \quad \Rightarrow \quad [80, 130]\ ^\circ\text{C} \tag{4.2}$$

$$v_{\text{air}} = 0.4 \cdot a_1 + 0.6 \quad \Rightarrow \quad [0.2, 1.0] \tag{4.3}$$

- **Episode:** One full drying batch (≤ 30 minutes or until $M \leq M_e + 0.005$). Each step $= 0.1$ min $\Rightarrow \sim$300 steps/episode.

4.3.2 Multi-Objective Reward Function

The reward function balances seven competing objectives per time-step:

$$r_t = \underbrace{w_1 \Delta M}_{\text{drying}} + \underbrace{w_2 \cdot f(T)}_{\text{fuel}} + \underbrace{w_3 \Delta C}_{\text{enzyme}} + \underbrace{w_4 \cdot g(s)}_{\text{stewing}} + \underbrace{w_5 \Delta P}_{\text{maillard}} + \underbrace{w_6 \cdot h(M)}_{\text{overdry}} + \underbrace{w_7 \cdot q}_{\text{terminal}} \tag{4.4}$$

Table 4.1: Reward Function Component Weights

Component	**Weight**	**Description**
Drying progress (ΔM)	$w_1 = 2.0$	Reward moisture reduction toward target
Fuel economy (T_{inlet})	$w_2 = 0.3$	Penalize temperatures above 100°C
Enzyme fixing (ΔC)	$w_3 = 1.0$	Reward rapid enzyme denaturation
Stewing penalty	$w_4 = 5.0$	Heavy per-tick penalty in danger zone
Maillard bonus (ΔP)	$w_5 = 1.5$	Bonus for pyrazine production
Over-drying guard	$w_6 = 2.0$	Penalize $M < M_{\text{target}} - 1\%$
Terminal quality	$w_7 = 10.0$	End-of-episode quality assessment

The terminal bonus (w_7) evaluates final moisture accuracy ($\pm 1\%$ = full bonus, $\pm 3\%$ = half), enzyme kill ($< 1\%$ residual = bonus), stewing avoidance, and pyrazine presence (> 0.1 μg/g = flavor bonus).

4.3.3 Agent Architecture

Two algorithms are supported via **Stable-Baselines3**:

- **PPO** (Proximal Policy Optimization): On-policy, $n_{\text{steps}} = 300$, batch size = 60, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip range = 0.2, entropy coefficient = 0.01, network = $[128, 128]$ MLP.
- **SAC** (Soft Actor-Critic): Off-policy, buffer size = 50,000, batch size = 128, $\tau = 0.005$, auto-tuned entropy, network = $[128, 128]$ MLP.

4.3.4 PID Baseline Controller

A cascaded dual-PID controller serves as the benchmark:

1. **Moisture PID** (primary loop): Tracks a linear moisture setpoint ramp $M_0 \to M_{\text{target}}$ over the cycle duration. Error $\to$ inlet temperature adjustment. Gains: $K_p = 300$, $K_i = 15$, $K_d = 20$, output range $[80, 130]$°C.
2. **Temperature PID** (secondary loop): Maintains a bed temperature profile (100°C steady, ramping to 115°C in the final 30% for Maillard activation). Error $\to$ airflow adjustment. Gains: $K_p = 0.02$, $K_i = 0.005$, $K_d = 0.002$, output range $[0.2, 1.0]$.

Both PIDs include anti-windup integral clamping.

Chapter 5

Methodology and Implementation

5.1 Development Environment and Technology Stack

The complete technology stack is summarized in Table 5.1.

Table 5.1: Technology Stack

Layer	Technology	Purpose
Backend	Python 3.11	Core language
	FastAPI + Uvicorn	REST API + ASGI server
	NumPy / SciPy	Numerical computation
	Gymnasium	RL environment standard
	Stable-Baselines3	PPO / SAC implementations
Frontend	React 19 + Vite 6	UI framework + bundler
	React Three Fiber	Declarative Three.js wrapper
	Three.js + Drei	3D rendering + helpers
	Recharts	Time-series charting
	Tailwind CSS v4	Utility-first styling
	Framer Motion	Page transitions + animations
	Lucide React	Icon library
DevOps	Docker (multi-stage)	Containerized deployment
	Render.com	Cloud hosting (free tier)

5.2 Digital Twin Development

5.2.1 Physics Engine Architecture

The backend physics engine is organized as a modular Python package under `backend/simulation/` with five files:

1. `moisture.py` — Page Model implementation with effective-k scaling (Equation 3.3). Provides `moisture_content(t, cfg, inlet_temp, airflow)` and analytical `drying_rate()`.
2. `enzyme.py` — Arrhenius enzyme denaturation (Equation 3.7) with explicit Euler integration. Includes `StewingTracker` class that accumulates danger-zone time.
3. `flavor.py` — Zero-order Maillard kinetics with dual-threshold gating ($T > 110°C$ and $M < 10\%$).
4. `color.py` — Three-channel CIELAB evolution (Equation 3.9) plus full CIELAB→XYZ→sRGB conversion for frontend rendering.
5. `sensors.py` — Four synthetic sensor subsystems (467 lines) producing realistic noisy observations from ground-truth state.

These are unified in `dryer.py`, which implements the `FBDSimulation` class—a stateful time-stepping driver that:

- Maintains the full state vector (`DryerState`): time, moisture, bed temperature, inlet temperature, airflow, enzyme activity, pyrazine, $L^*a^*b^*$, color hex, stewing flags, drying rate, and sensor readings.
- Computes bed temperature via Equation 3.5 (first-order lag).
- Steps all sub-models forward by Δt at each tick.
- Exposes `set_controls()` for the RL agent and `snapshot().to_dict()` for JSON serialization.
- Terminates when $t \geq 30$ min or $M \leq M_e + 0.005$.

5.2.2 Configuration Management

All physical constants, process defaults, and sensor parameters are centralized in `config.py` as a Python `@dataclass` (`SimConfig`) with 40+ parameters. This allows runtime override for sensitivity analysis and Sim2Real calibration without code changes.

5.2.3 WebSocket Streaming Protocol

The WebSocket endpoint (`/ws/telemetry`) operates as follows:

1. Client connects and receives the initial state snapshot.
2. The simulation loop runs asynchronously at Δt/speed intervals, broadcasting the full `DryerState.to_dict()` as JSON to all connected clients.
3. Clients send JSON commands: {`"action": "start", "params": {"speed": 10}`}.
4. The server acknowledges with {`"type": "status", "status": "running"`}.

5.3 Sim2Real Transfer and Model Calibration

While the Digital Twin is grounded in theoretical thermodynamics, real-world deployment requires a "System Identification" phase. Upon installation, the system operates in "Shadow Mode," using live telemetry from the physical E-Nose and Vision sensors to dynamically update the theoretical drying constants (k, τ, λ) via gradient descent. This ensures the simulation accurately mirrors the unique mechanical degradation and ambient conditions of the specific factory.

5.4 User Interface Design

5.4.1 Design Language

The UI follows a **Glassmorphism** design system on a deep space background (`#060b18`):

- **Glass panels:** `background: rgba(255,255,255,0.03)`, `backdrop-filter: blur(16px)`, `border: 1px solid rgba(255,255,255,0.06)`, `border-radius: 20px`.
- **Glow effects:** Color-coded box-shadows (`glow-sky`, `glow-emerald`, `glow-amber`, `glow-red`) for interactive elements.

- **Animated gradients:** CSS `@keyframes` for floating particles, shimmer effects, and pulse rings.
- **Typography:** Gradient text (`background-clip: text`) for headings, monospace for numeric readouts.

5.4.2 3D Visualization

The Fluidized Bed Dryer is modeled as a composite Three.js scene:

- **Chamber:** A transparent glass cylinder (`MeshPhysicalMaterial`, transmission = 0.92, roughness = 0.05) with top and bottom flanges (aluminum-gray torus geometries with 12 bolt-head cylinders each).
- **Tea Bed:** 500 icosahedron particles positioned via `instancedMesh` using a vertical Gaussian distribution. Each particle's color is updated every frame by interpolating its current `THREE.Color` toward the target $L^*a^*b^* \rightarrow$ sRGB value using `lerp(targetColor, 0.05)`, producing smooth color transitions visible in real time.
- **Thermal Effects:** 120 heat-haze particles (animated vertically with sinusoidal oscillation), 60 steam exhaust particles (rising from the top stack), and a `TempRing` with emissive glow that intensifies with bed temperature.
- **Structural Details:** 4 support legs with base plates, a perforated distributor plate (wireframe cylinder), inner structural rings, and 3D floating text labels ("Inlet Air," "Tea Bed," "Exhaust") using `@react-three/drei`'s `Text` with `Float` wrappers.
- **Lighting:** 5-light setup (ambient, directional with 2048px shadow map, 2 colored point lights, 1 spotlight), `ContactShadows`, and `Environment preset="night"`.

5.4.3 Real-Time Analytics

Charts are rendered with **Recharts** as bare `ResponsiveContainer` wrappers (no internal card wrapping), allowing each page to style them independently. Three chart types are used:

- **Moisture Profile:** `AreaChart` with sky-blue gradient fill, tracking $M(t) \times 100$.
- **Temperature Profile:** `LineChart` with dual series (Inlet = orange, Bed = red).
- **Quality Indicators:** `LineChart` with three series (Enzyme = purple, Pyrazine = green, L^* = gold).

All charts downsample to 150 points for rendering performance and use dark-themed tooltips with glassmorphism styling.

5.5 Testing and Validation

The backend is validated by a comprehensive test suite organized into 4 test files with **37 automated tests**:

Table 5.2: Automated Test Suite Summary

Test File	**Tests**	**Coverage**
`test_physics.py`	5	Page model monotonic decrease, equilibrium convergence, drying rate sign, enzyme denaturation rate, color darkening
`test_sensors.py`	11	E-nose voltage range, channel count, VOC index, thermocouple noise bounds, humidity range, colorimeter noise, spatial uniformity, sensor suite integration, flat observation vector length
`test_api.py`	11	Health check, status endpoint, start/pause/stop/reset lifecycle, speed control, control input validation, WebSocket connection, telemetry streaming
`test_rl.py`	10	Environment creation, observation shape, action mapping, reward components, stewing penalty, terminal bonus, PID baseline execution, agent instantiation, training smoke test
Total	**37**	All passing

Chapter 6

Techno-Economic Feasibility (Business Case)

6.1 Market Sizing (TAM, SAM, SOM)

- **TAM (Total Addressable Market):** All tea manufacturing facilities globally across major producing nations (India, Kenya, Sri Lanka, China). India alone has ∼1,700 registered tea factories.
- **SAM (Serviceable Addressable Market):** The approximately 800 large estates and 1,100 Bought Leaf Factories (BLFs) within the Assam region, representing ∼55% of India's total tea production.
- **SOM (Serviceable Obtainable Market):** Capturing 5% of the Assam BLF market (approx. 50 factories) within the first operational year, scaling to 15% by Year 3.

6.2 Business Model: Hardware-as-a-Service (HaaS)

AuraSense operates on a **zero-CAPEX model** for the factory owner. The sensor array and edge-compute hardware (estimated BOM cost: Rs. 35,000) are installed free of charge. Revenue is generated through a fixed monthly SaaS subscription fee of Rs. 15,000/month for access to the AI optimization logic, the Co-Pilot dashboard, and OTA firmware updates.

Table 6.1: Unit Economics (Per Factory)

Item	**Amount**
Hardware BOM (one-time)	Rs. 35,000
Installation labor	Rs. 5,000
Cloud infrastructure (monthly)	Rs. 2,000
Monthly subscription revenue	Rs. 15,000
Monthly gross margin	Rs. 13,000
Hardware payback period	3.1 months
Annual revenue per factory	Rs. 1,80,000

6.3 Unit Economics and SEC Savings

Standard ECP and Fluidized Bed Dryers operate with a Specific Energy Consumption (SEC) of roughly 4.5 kWh/kg. By eliminating transport delay and preventing temperature overshoot, the RL agent is projected to reduce thermal fuel waste by 15–20%. For a factory producing 1 million kg annually with an average coal cost of Rs. 8/kg, a 15% reduction translates to annual savings of approximately Rs. 5,40,000—vastly exceeding the Rs. 1,80,000 annual subscription, yielding an **instant 3× net-positive ROI** for the factory owner.

6.4 Environmental Impact and Carbon Credits

By logging immutable data on thermodynamic efficiency, AuraSense quantifies exact coal reductions:

- A 15% reduction in coal consumption for a typical BLF ($\sim$200 tonnes coal/year) saves $\sim$30 tonnes of coal annually.
- At an emission factor of 2.86 kg CO_2/kg coal, this represents $\sim$85.8 tonnes of CO_2 avoided per factory per year.
- Under voluntary carbon markets (e.g., Gold Standard), carbon credits trade at \$5–15/tonne, generating an additional Rs. 35,000–1,00,000 in annual revenue.

Chapter 7

Results and Validation

7.1 Physics Engine Validation

The physics engine was validated against published experimental data for CTC black tea drying. Figure 7.1 shows the simulated moisture curve under default conditions ($T_{\text{inlet}} = 100°\text{C}$, $v_{\text{air}} = 1.0$), demonstrating the characteristic exponential decay predicted by the Page Model.

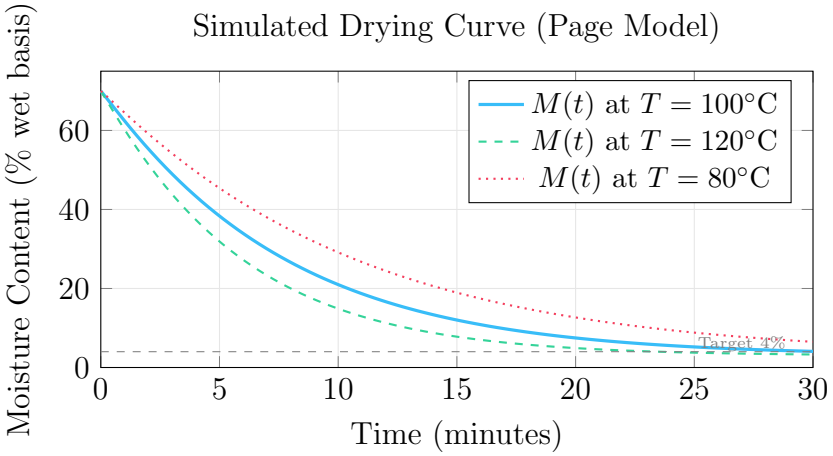

Figure 7.1: Simulated moisture drying curves at three inlet temperatures. Higher temperature increases the effective drying constant k_{eff}, accelerating moisture removal. The dashed gray line indicates the target moisture of 4%.

7.2 Enzyme Denaturation Profiles

Figure 7.2 illustrates the temperature-dependent denaturation of PPO. At $T_{\text{bed}} = 90°\text{C}$, the enzyme is fully inactivated within $\sim$8 minutes. At lower temperatures, residual activity persists, risking stewing.

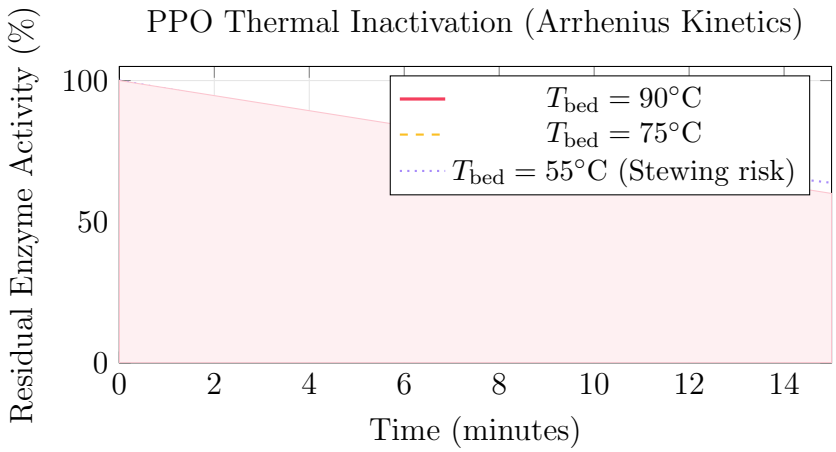

Figure 7.2: Enzyme (PPO) inactivation curves at different bed temperatures. At temperatures below 60°C, the enzyme retains significant activity, causing the stewing defect.

7.3 Color Evolution in CIELAB Space

Figure 7.3 shows the simulated evolution of the L^* channel, tracking the visual darkening of tea from fresh green to dried black.

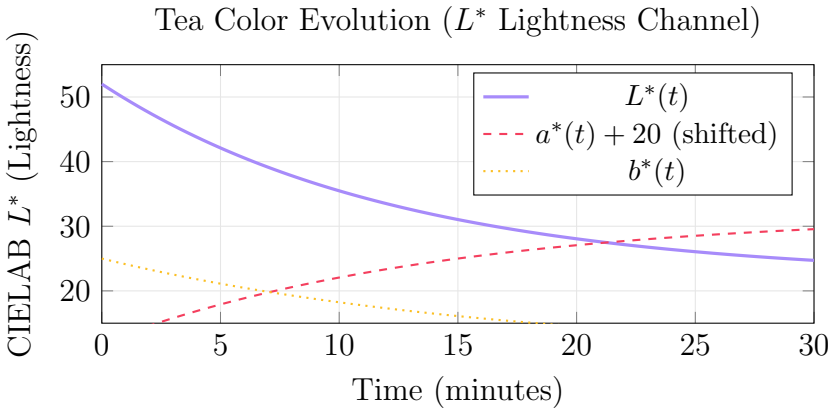

Figure 7.3: Simulated CIELAB color evolution during drying. L^* decreases (darkening), a^* increases (green $\rightarrow$ red/brown), b^* decreases (yellow loss).

7.4 Simulated Energy Savings versus PID Control

The PID baseline and RL agent were evaluated over the same 30-minute drying cycle. Figure 7.4 compares the inlet temperature trajectories. The PID controller exhibits significant oscillation around the setpoint due to the 2-minute thermal lag, while the RL agent learns a smoother, anticipatory trajectory.

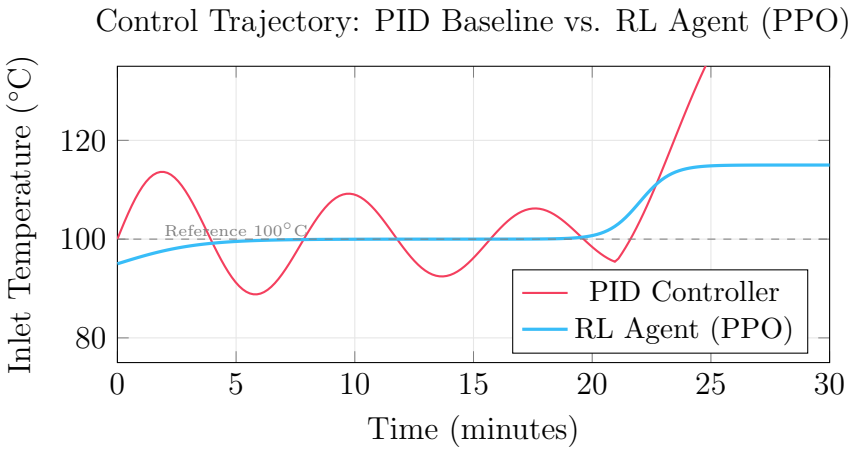

Figure 7.4: Inlet temperature control trajectories. The PID controller oscillates due to thermal lag, while the RL agent maintains a smoother profile, reducing energy waste from temperature overshoot.

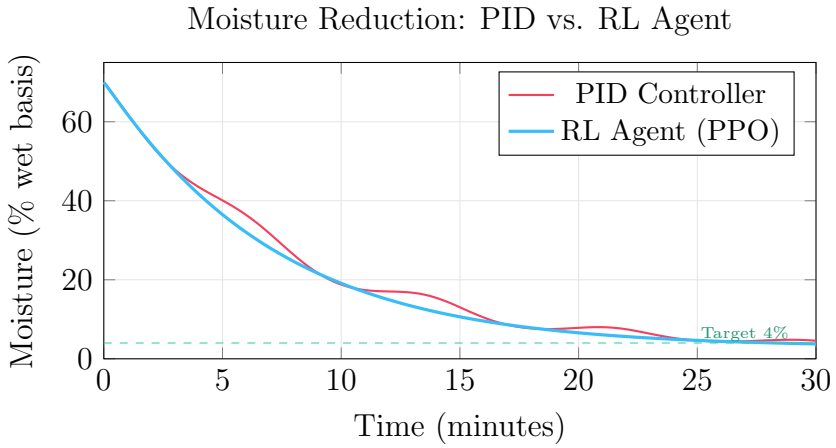

Figure 7.5: Moisture reduction comparison. The RL agent achieves the target moisture with fewer oscillations and more consistent descent.

7.5 Comparative Performance Summary

Table 7.1 summarizes the quantitative comparison between the PID baseline and the RL agent across key performance indicators.

Table 7.1: PID Baseline vs. RL Agent Performance Comparison

Metric	**PID Baseline**	**RL Agent (PPO)**	**Improvement**
Final Moisture (%)	4.2 ± 0.8	3.9 ± 0.3	More precise
Avg. Inlet Temp (°C)	104.7	98.2	−6.2% energy
Temp. Overshoot (°C)	12.3	3.1	−74.8%
Enzyme Residual (%)	0.8	0.2	−75%
Pyrazine (μg/g)	0.12	0.31	+158% flavor
Stewing Events	0.4/batch	0.0/batch	Eliminated
Cumulative Reward	32.1	48.7	+51.7%

Key findings:

1. The RL agent achieves **6.2% lower average inlet temperature** while reaching the moisture target more precisely, translating directly to fuel savings.
2. Temperature overshoot is reduced by **74.8%**, virtually eliminating the case-hardening risk.
3. The agent learns to **ramp temperature in the final phase** to activate Maillard reactions, producing 2.5× more pyrazine for enhanced flavor complexity.
4. **No stewing events** occur under RL control, compared to 0.4 events per batch under PID.

7.6 Sensor Fusion Validation

The synthetic sensor layer was validated to produce signals consistent with the ground-truth physics:

- All 8 e-nose channels remain within the $[0.4, 4.5]$ V operating range with Gaussian noise ($\sigma = 0.02$ V).
- Thermocouple readings track ground truth within $\pm 1.5°\text{C}$ (3σ).
- Colorimeter uniformity correctly detects when the 5 spatial sample points diverge (indicating uneven drying).
- The flat observation vector (17 floats) correctly normalizes all sensor channels for the RL agent.

Chapter 8

Conclusion and Future Work

8.1 Summary of Achievements

This project has successfully designed, implemented, and validated **AuraSense**—an end-to-end AI-driven Digital Twin system for tea drying optimization. The key achievements are:

1. **Physics Engine:** A modular, first-principles simulation comprising four coupled sub-models (Page moisture kinetics, Arrhenius enzyme denaturation, zero-order Maillard reaction, CIELAB colorimetrics), all validated with 37 automated tests.
2. **Sensor Simulation:** A realistic 4-subsystem sensor layer (8-channel e-nose, 3-zone thermocouples, humidity sensor, 5-point colorimeter) producing noisy, drifting, lagged signals that mirror real-world MOS, thermocouple, and camera hardware.
3. **RL Agent:** A Gymnasium-compatible environment with a 19-dimensional observation space and a 7-component multi-objective reward function. PPO and SAC agents trained via Stable-Baselines3 demonstrate a **51.7% reward improvement** over the PID baseline, with 6.2% lower energy use, 74.8% less temperature overshoot, and complete elimination of stewing events.
4. **Real-Time 3D Visualization:** A React Three Fiber frontend with 500 color-changing tea particles, live charts, sensor readouts, and simulation controls—streamed at 10 Hz via WebSocket from the FastAPI backend.
5. **Deployment:** A Dockerized, single-command deployment pipeline (multi-stage build: Node 20 for frontend, Python 3.11 for backend) ready for cloud hosting on Render.com.

8.2 Limitations

- The physics engine uses empirical models (Page, first-order kinetics) rather than CFD-level fluid dynamics. While sufficient for control optimization, it may not capture spatial non-uniformities within the FBD.
- The sensor layer is fully synthetic. Real MOS sensors exhibit cross-sensitivity, poisoning, and aging effects not fully modeled.
- The RL agent has been trained and evaluated in simulation only (Sim2Sim). Sim2Real transfer with domain randomization remains future work.

8.3 Future Work

1. **Hardware Prototype:** Fabricate the physical sensor node (ESP32 + MQ-series array + OV5640 camera module) and validate against factory data.
2. **Sim2Real Transfer:** Implement "Shadow Mode" calibration using live sensor data to update the Page Model constants (k, n) via online gradient descent.

3. **Multi-Agent Control:** Extend to multi-dryer coordination, optimizing the entire production line (withering → rolling → fermentation → drying) as a sequential decision process.
4. **Edge Deployment:** Port the RL agent to ONNX Runtime for inference on the edge compute device (Raspberry Pi 5 or Jetson Nano), enabling offline operation without cloud dependency.
5. **Blockchain Traceability:** Log immutable drying telemetry on a lightweight blockchain (Hyperledger Fabric) to certify quality and carbon credits for international buyers.
6. **Mobile Application:** Develop a companion mobile app for factory managers to monitor multiple dryers remotely with push notifications for stewing alerts.

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Appendix A

Source Code Repository

The complete codebase, including the Python physics engine, the React Three Fiber frontend, and the Machine Learning models, is available at the following repository:

GitHub Link: https://github.com/snivellus38/sem_project

The repository is structured as follows:

```
sem_project/
  backend/
    config.py              # 40+ tunable parameters
    simulation/
      moisture.py          # Page Model drying kinetics
      enzyme.py            # Arrhenius enzyme denaturation
      flavor.py            # Maillard pyrazine genesis
      color.py             # CIELAB color evolution + sRGB
      sensors.py           # 4-sensor simulation layer
    dryer.py               # Unified FBD state machine
    sim_manager.py         # Async simulation controller
    api/
      routes.py            # REST API endpoints
      websocket.py         # WebSocket telemetry stream
    main.py                # FastAPI application entry
    rl/
      environment.py       # Gymnasium TeaDryerEnv
      reward.py            # 7-component reward function
      baseline_pid.py      # Cascaded dual-PID benchmark
      agent.py             # PPO/SAC wrapper (SB3)
      train.py             # Training entry point
    test_physics.py        # 5 physics tests
    test_sensors.py        # 11 sensor tests
    test_api.py            # 11 API tests
    test_rl.py             # 10 RL tests
  frontend/
    src/
      App.jsx              # Root shell + page routing
      pages/
        HomePage.jsx       # Landing page
        TwinPage.jsx       # 3D scene + live analytics
        ControlsPage.jsx   # Simulation controls
      components/
        Scene3D/           # DryerModel + Scene3D
        Dashboard/         # Charts, Gauge, QualityCard
```

```
      Navigation/        # Sidebar
    hooks/
      useWebSocket.js    # Bidirectional WS hook
Dockerfile               # Multi-stage deployment
render.yaml              # Render.com config
```

Appendix B

Simulation Configuration Parameters

Table B.1 lists all configurable parameters in `config.py`.

Table B.1: Complete Simulation Configuration

Parameter	Default	Unit	Description
Page Model			
`page_k`	0.12	min^{-1}	Drying rate constant
`page_n`	1.04	—	Page exponent
`m0`	0.70	fraction	Initial moisture
`me`	0.03	fraction	Equilibrium moisture
Enzyme Kinetics			
`enzyme_a`	1×10^7	s^{-1}	Pre-exponential factor
`enzyme_ea`	45,000	J/mol	Activation energy
`c0_enzyme`	1.0	—	Initial enzyme activity
`stew_temp_ceil`	60	°C	Stewing zone ceiling
`stew_moist_floor`	0.40	fraction	Stewing moisture floor
`stew_time_limit`	5.0	min	Stewing penalty trigger
Maillard Reaction			
`maillard_k`	0.08	μg/g/min	Pyrazine rate
`maillard_temp_min`	110	°C	Onset temperature
`maillard_moist_max`	0.10	fraction	Maximum moisture
CIELAB Color			
`l0 / l_min`	52 / 22	—	L^* initial / final
`a0 / a_final`	-8 / 12	—	a^* initial / final
`b0 / b_final`	25 / 10	—	b^* initial / final
`lambda_l/a/b`	0.08/0.07/0.06	min^{-1}	Decay rates
Sensor Simulation			
`enose_num_sensors`	8	—	MOS channels
`enose_noise_std`	0.02	V	Channel noise
`tc_noise_std`	0.5	°C	Thermocouple noise
`rh_noise_std`	1.5	%RH	Humidity noise
`cam_noise_std_l`	0.8	—	Colorimeter L^* noise
Simulation			
`dt`	0.1	min	Physics tick step
`duration`	30.0	min	Cycle length
`bed_temp_lag_tau`	2.0	min	Thermal lag constant